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INTRODUCTION

A **log** is a recorded entry of events generated by systems, applications, or network devices that shows what happened, when it happened, and sometimes who or what caused it. **Log monitoring and log analysis** is the process of continuously collecting, reviewing, and analysing these logs to detect errors, security incidents, suspicious activities, and performance issues. It helps organizations identify threats early, investigate incidents, ensure system health, and meet compliance requirements by providing visibility into system behaviour and user actions.

OBJECTIVE

This project's main goal is to how to read and monitor logs and analyse logs from any operating system to Linux operating system using Wazuh SIEM Tool. With the user agent endpoints of Wazuh SIEM you can almost monitor and analyse logs in Linux. We will be going from installation of both main component installation in Linux OS and endpoint in Windows OS. This project helps in improving knowledge of both Linux and Windows.

PREREQUISITE

HARDWARE

Processor: 64-bit Intel, AMD, or ARM Linux Processor

Ram: 4GB or Above

SOFTWARE

- Kali Linux
  Operating system to monitor the receiving logs. It doesn't have to be Kali Linux but Wazuh SIEM is UNIX-based operating server. Other than endpoints, you can't use windows to monitor using Wazuh.

- Windows 8.1
  Operating system where log is sending from. (Refrain from using windows 7 as it is way too outdated and it waste your time more)

- Virtual Box
  Where both the Kali Linux and Windows 8.1 is used to run virtually as it is used to create and manage virtual machines.

- Wazuh (Server, Dashboard and Indexer)
  The SIEM tool used to monitor and analyse Logs.

- Wazuh (Endpoint Agent)
  The Endpoint used to send logs to server

PROCEDURES

INSTALLATION OF MONITORING TOOLS

WAZUH (SERVER, DASHBOARD AND INDEXER)

Step 1: Open Kali Linux terminal and type:
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a

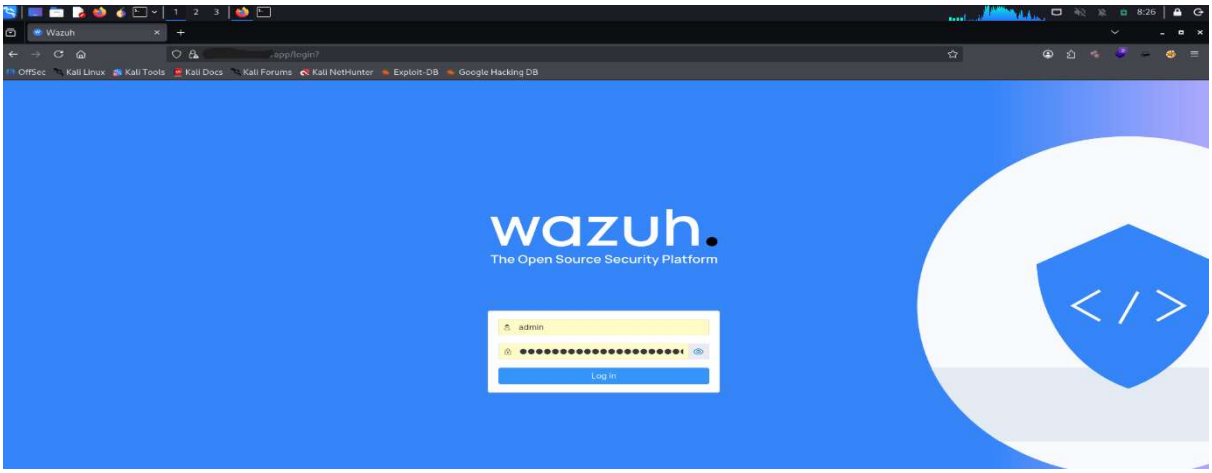
```
┌──(kali㉿kali)-[~]
└─$ curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
```

Step 2: After some time, it will show the summary with username and password

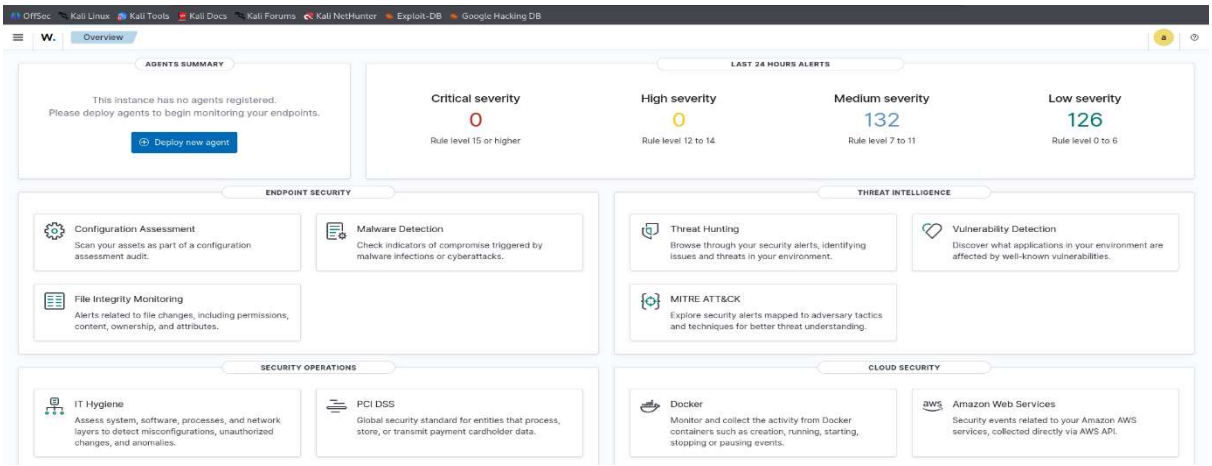
```
03/02/2026 07:32:03 INFO: --- Summary ---
03/02/2026 07:32:03 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password: ArlXWhW+gfe51JekAchUt9P4iUDALHo+
03/02/2026 07:32:03 INFO: Installation finished.
```

Step 3: Now you can log into the Wazuh dashboard. Just open any browser and type:
https://<MACHINE_IP_ADDRESS>:443

(Replace <MACHINE_IP_ADDRESS> with your IP address and type ‘https’ not ‘http’. When you access the Wazuh dashboard for the first time, the browser shows a warning message stating that the certificate was not issued by a trusted authority. Just accept it and load the page)

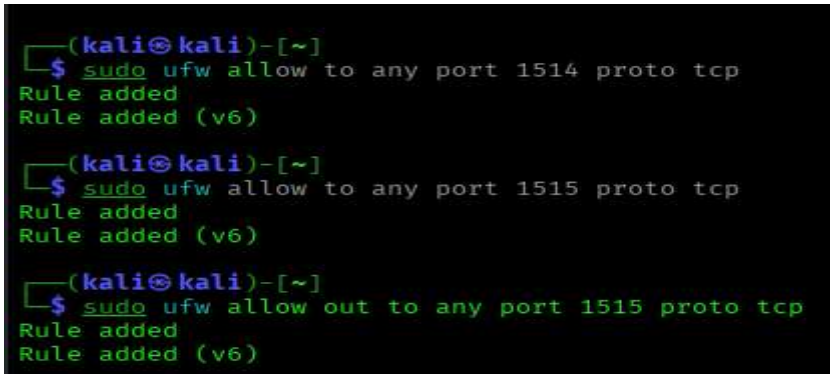


Step 4: After entering username and password it will show like this:



Step 5: Type these three codes in kali linux terminal:

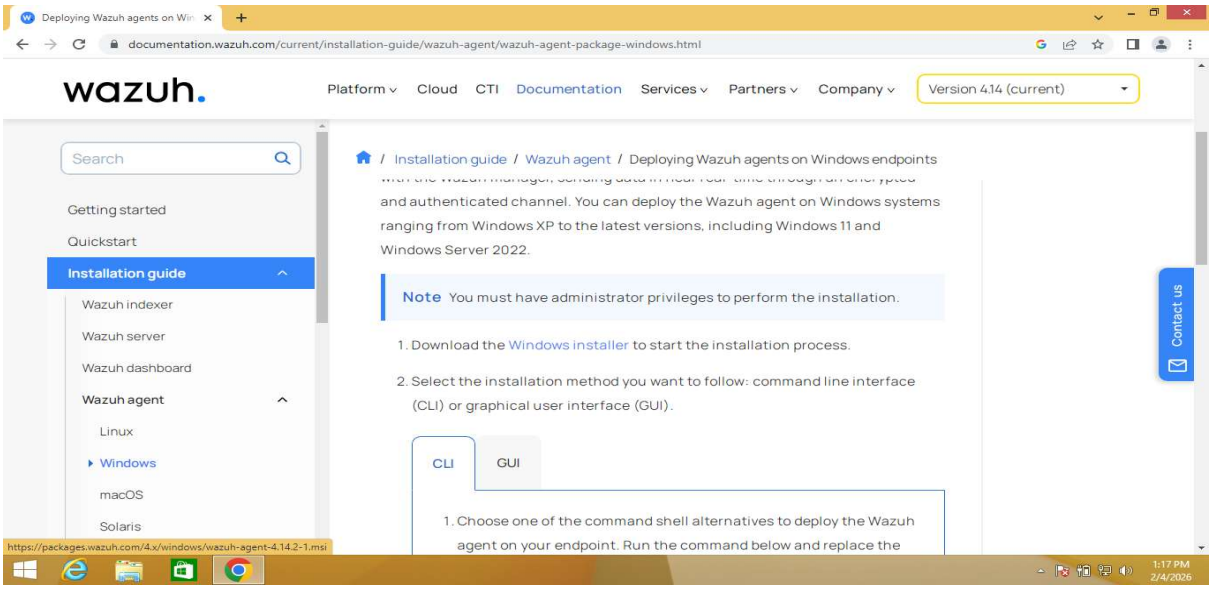
sudo ufw allow to any port 1514 proto tcp
sudo ufw allow out to any port 1515 proto tcp
sudo ufw allow to any port 1515 proto tcp



if ufw is not installed use ‘sudo apt-get install ufw’.

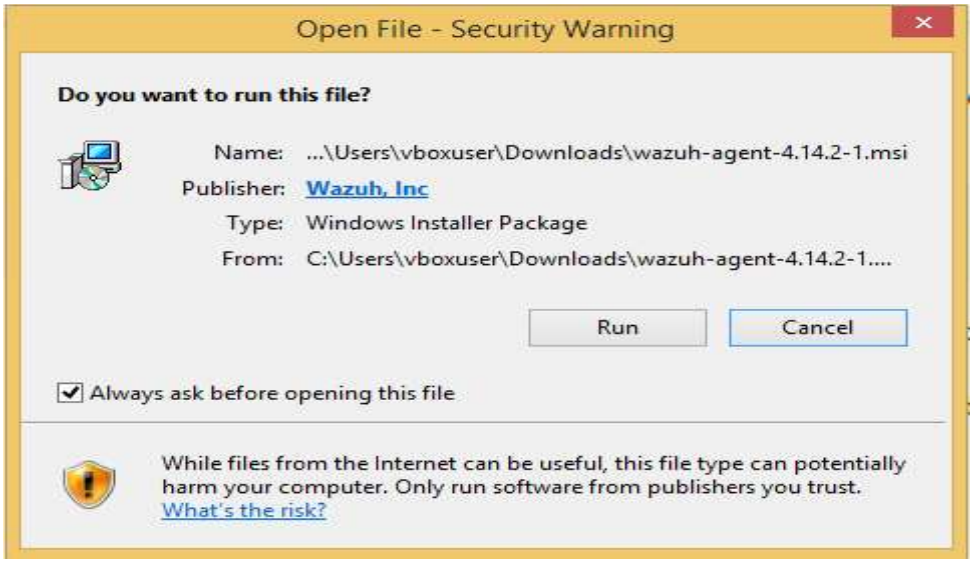
WAZUH (AGENT ENDPOINT)

Step 1: Start Windows and open a browser and search for the given link: https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-windows.html

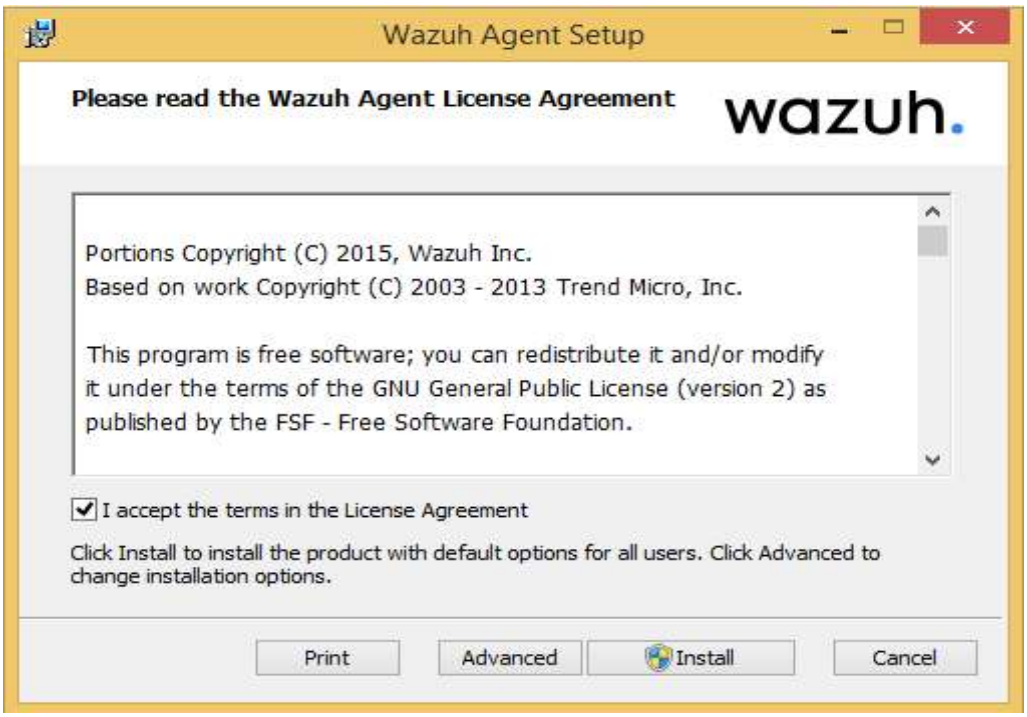


Step 2: Click on Windows Installer and windows installer for wazuh agent will download. Click on it after download.

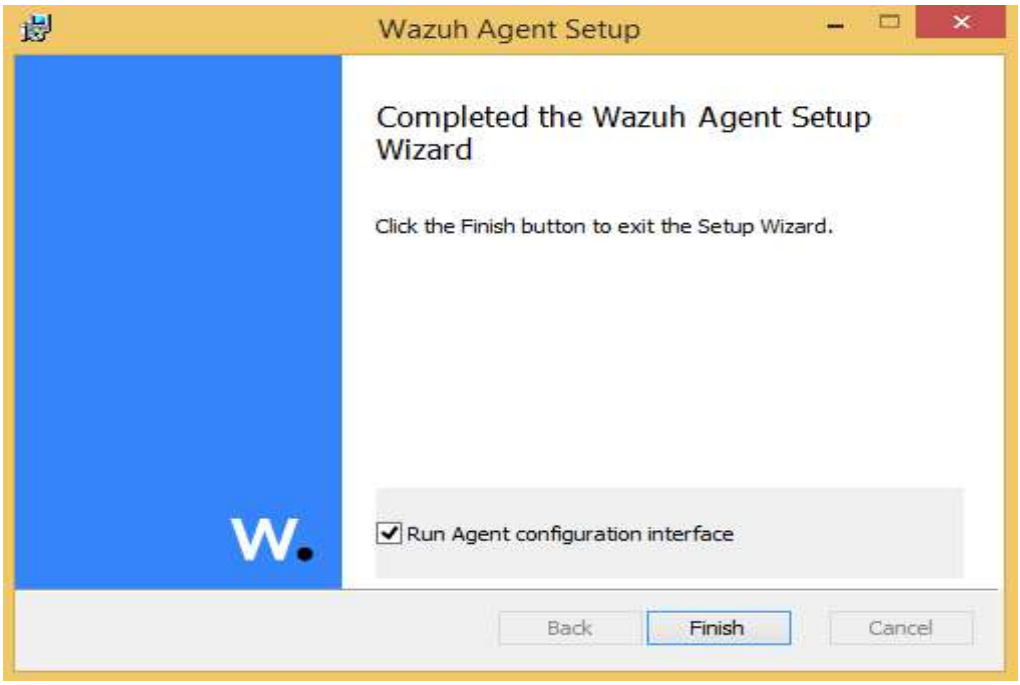
Step 3: Click on install and click on Run.



Step 4: Accept the terms and conditions and click on install. Click on Yes when a prompt box pops up.

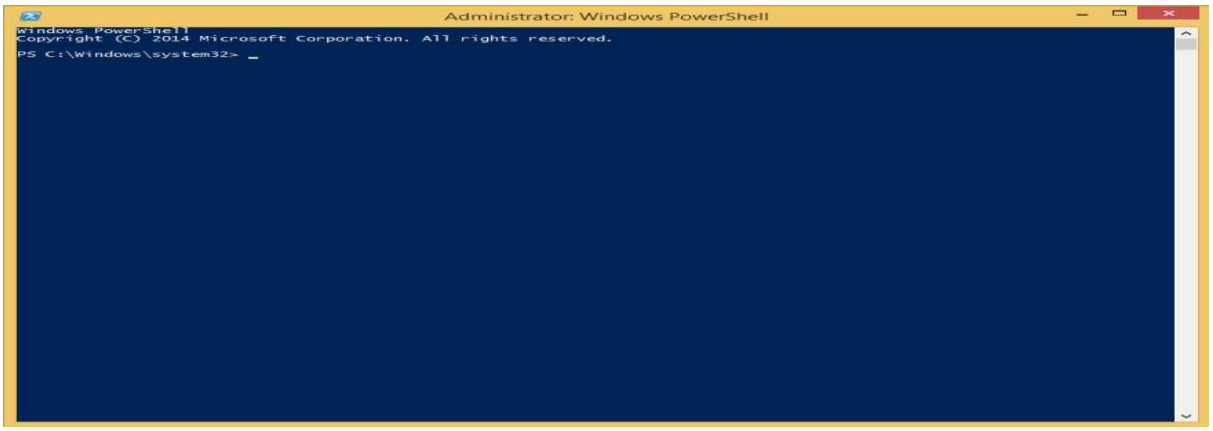


Step 5: After Installation, select 'Run Agent configuration interface' and click finish. Click on Yes when a prompt box pops up.



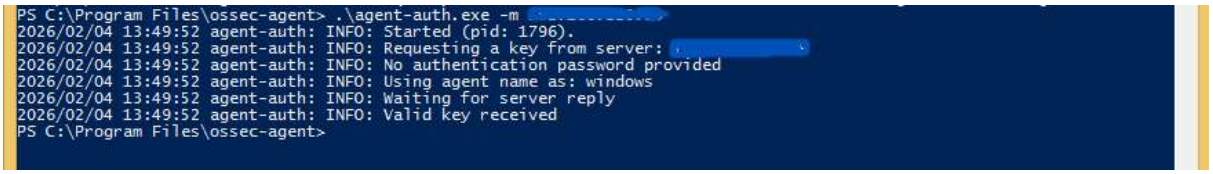
Step 6: After clicking finish, The GUI of Wazuh Agent will open. Type Your Kali Linux IP address (Where Wazuh server is installed) and click on Save.

Step 7: To get the Authentication Key, Click on Start, search for Powershell and Left-click it and select Run as Administrator.

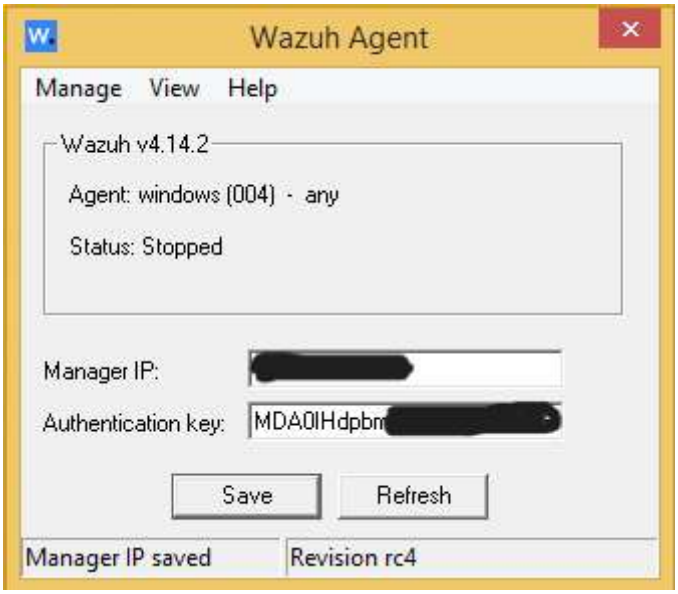


Step 8: Type ‘cd ‘C:\Program Files\ossec-agent’’ (folder where the Wazuh agent files are located) to change directory to it and enter.

Step 9: Type ‘.\agent-auth.exe -m <MANAGER_IP_ADDRESS>’, replace MANAGER_IP_ADDRESS with Kali Linux IP address, and enter. This will get you the Authentication key from Kali Linux.

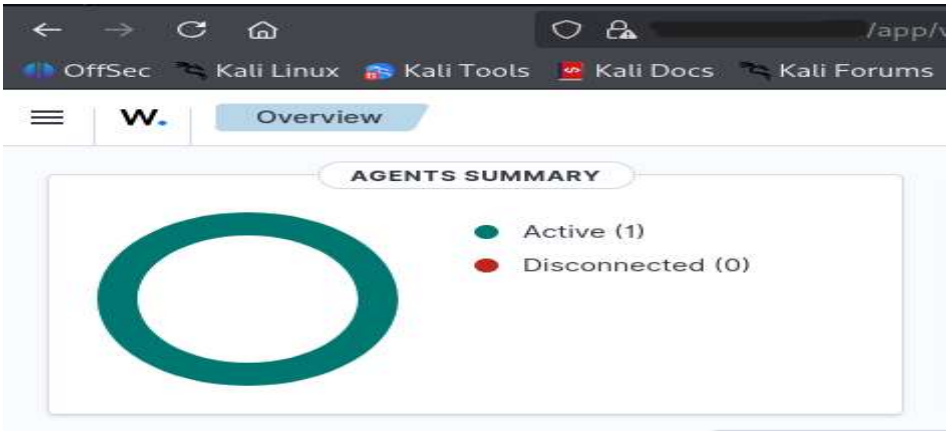


Step 10: Click on Refresh to see the Authentication Key to appear

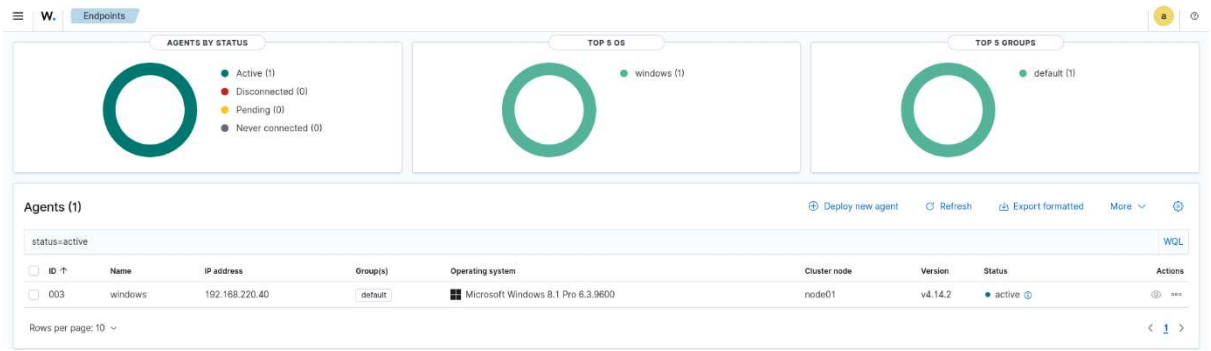


Step 11: Click on Manage and Select Restart, this will restart the Agent.

After all this Steps, you can see that the agent is registered in the server when you see this

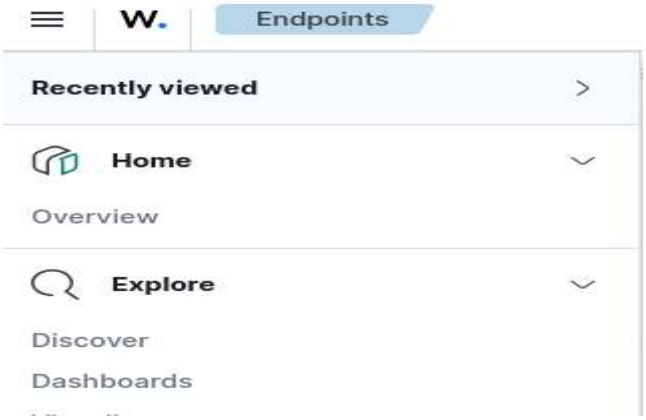


Clicking on Active will show the agents added in the manager and other agents' details.

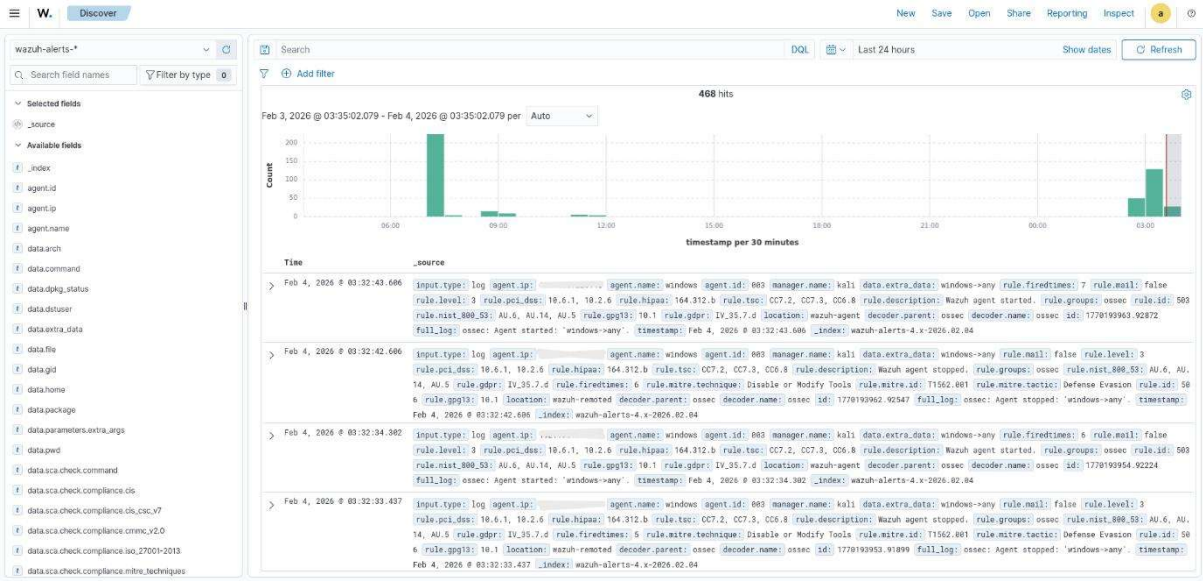


LOG ANALYSIS

To analyse the log received from the Windows agent, open the Wazuh server in the Kali Linux and click on hamburger menu on top left corner of the site, expand Explore and click on Discover.



It will open the Discover tab where you can filter or search for the log data based on agent name, agent ip address, status code, …



CHALLENGES

Installation of Wazuh in Kali Linux was easy but installation of agent in Windows was not easy as thought. Installing agent in windows using Command Prompt was not working every time I tried. Installation using GUI was easy but if you don't know how to get the Authentication Key it is not possible through that. After Installing everything the log was not receiving on the manager due to firewall issue in Kali Linux, where it was solved using command 'ufw'.

CONCLUSION

The installation and configuration of the Wazuh main components and the Wazuh agent were completed successfully. The agent was able to establish secure communication with the Wazuh manager, and system logs from the agent were effectively collected and forwarded. These logs were monitored, analyzed, and processed by the Wazuh manager, generating meaningful security events and alerts visible on the dashboard. This confirms that the Wazuh SIEM setup is functioning correctly and is capable of centralized log monitoring and security analysis.

REFERENCES

Wazuh Main Component Installation:
https://documentation.wazuh.com/current/quickstart.html

Wazuh Agent Installation:
https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-windows.html
https://www.youtube.com/watch?v=vNnijRi-pYA

Other Helps:
https://chatgpt.com